Fiber Channel Validation

Graph Documentation — Model vs. Theory

opto-sim

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1 Overview

This document explains every panel in the four validation figures produced by `analysis/validation/`. Each impairment is validated against its source literature. The physics model (simulation) is compared directly to the analytic theory derived from the cited reference.

The figures are:

1. Fiber Attenuation — `val_attenuation--seed42.png`
2. Fiber Birefringence — `val_birefringence--seed42.png`
3. Chromatic Dispersion — `val_cd--seed42.png`
4. Polarization Mode Dispersion — `val_pmd_dgd--seed42.png`

2 Fiber Attenuation

2.1 Source Literature

Keiser, G., *Optical Fiber Communications*, 5th ed., McGraw-Hill, 2015. Equation (3.6).

2.2 Theory

Optical power decays exponentially with fiber length. Per Keiser [1] Eq. (3.6):

$$P(L) = P_0 \cdot 10^{-\alpha L/10} \tag{1}$$

where α is the attenuation coefficient in $\mathrm{dB\,km^{-1}}$. For SMF-28 at $\lambda = 1550\,\mathrm{nm}$, $\alpha = 0.182\,\mathrm{dB\,km^{-1}}$.

2.3 Model

`apply_attenuation(E, L_km, attenuation_factor)` multiplies the complex-envelope field by

$$\sqrt{10^{-\alpha L/10}} \tag{2}$$

preserving the field phase while reducing amplitude. This is equivalent to applying Eq. (1) at the field level, since $P = \langle |E|^2 \rangle$.

2.4 Panel Descriptions

Panel A: Power decay (Keiser Eq. 3.6)

- **Black line:** Analytic theory — $P(L) = P_0 \cdot 10^{-0.182\,L/10}$ on a logarithmic y-axis.
- **Red dots:** Measured output power from `apply_attenuation` at each distance.
- **Finding:** The red dots lie exactly on the black line across 5 orders of magnitude (10^0 to 10^{-4}). The simulation reproduces the exponential decay predicted by Keiser Eq. (3.6) with no deviation.

Panel B: Loss vs. distance

- **Blue dots:** Measured loss in dB at each distance: $-10\log_{10}(P_{\text{out}}/P_{\text{in}})$.
- **Black dashed line:** Nominal $\alpha = 0.182\,\text{dB}\,\text{km}^{-1}$ (linear in dB).
- **Red solid line:** Linear fit: $\alpha_{\text{fit}} = 0.182\,000\,\text{dB}\,\text{km}^{-1}$ ($R^2 = 1.0$).
- **Finding:** The dB loss is perfectly linear with distance, confirming a constant attenuation coefficient. The fitted α matches the nominal value exactly.

Panel C: Residual

- **Red line:** Percentage error $|P_{\text{measured}} - P_{\text{theory}}|/P_{\text{theory}} \times 100$ at each distance.
- **Finding:** The maximum relative error is $\sim 3.2 \times 10^{-14}\,\%$, which is floating-point precision. The simulation is exact to machine accuracy.

Panel D: SMF-28 attenuation spectrum (Keiser Fig. 3.2)

- **Gray line with markers:** Typical SMF-28 attenuation at standard wavelengths (800, 1000, 1200, 1310, 1385, 1450, 1550, 1625 nm).
- **Red dashed lines:** Markers at $\lambda = 1550\,\text{nm}$ and $\alpha = 0.182\,\text{dB}\,\text{km}^{-1}$.
- **Finding:** The operating point is correctly positioned on the standard SMF-28 attenuation spectrum. Attenuation decreases from $\sim 2.5\,\text{dB}\,\text{km}^{-1}$ at 800 nm to $0.182\,\text{dB}\,\text{km}^{-1}$ at 1550 nm, dominated by Rayleigh scattering at shorter wavelengths.

Panel E: Alpha consistency across distance

- **Blue dots:** Attenuation coefficient extracted independently at each distance: $-10\log_{10}(P_{\text{out}}/P_{\text{in}})/L$.
- **Red dashed line:** Nominal $\alpha = 0.182\,\text{dB}\,\text{km}^{-1}$.
- **Finding:** The extracted α is constant at $0.182\,\text{dB}\,\text{km}^{-1}$ for all distances from 5 km to 200 km. No distance-dependent drift confirms uniform application of Keiser Eq. (3.6).

Panel F: Validation summary

Tabulated values of P_{measured}, P_{theory}, and error at 0, 20, 50, 100, 150, and 200 km. All errors are at floating-point precision ($\sim 10^{-14}\,\%$).

3 Fiber Birefringence

3.1 Source Literature

1. Agrawal, G. P., *Fiber-Optic Communication Systems*, 5th ed., Wiley, 2021. Equation (4.1.2).
2. Ulrich, R. et al., "Bending-induced birefringence in single-mode fibers," *Opt. Lett.*, vol. 5, no. 6, pp. 273–275, 1980.

3.2 Theory

Birefringence introduces a Jones matrix that applies different phase shifts to the two polarization components. Per Agrawal [2] Eq. (4.1.2):

$$\mathbf{J} = \begin{pmatrix} e^{j\,\Delta\beta\,L/2} & 0 \\ 0 & e^{-j\,\Delta\beta\,L/2} \end{pmatrix} \tag{3}$$

where the differential propagation constant is

$$\Delta\beta = \frac{2\pi\,\Delta n}{\lambda} \tag{4}$$

and Δn is the birefringence (refractive-index difference between the two axes).

The birefringence Δn depends on temperature and bend radius per the Ulrich model [3]:

$$\Delta n = \Delta n_{T_0} + c_T\,(T - 25) + 0.135\left(\frac{r_{\text{fiber}}}{R}\right)^2 \tag{5}$$

where Δn_{T_0} is the intrinsic birefringence at $T_0 = 25\,°\mathrm{C}$, $c_T = -5 \times 10^{-7}\,°\mathrm{C}^{-1}$ is the temperature coefficient, r_{fiber} is the fiber cladding radius, and R is the bend radius.

The beat length is

$$L_B = \frac{\lambda}{\Delta n} \tag{6}$$

3.3 Model

`apply_birefringence(E, L, wavelength, temperature, bend_radius)` constructs the Jones matrix (3) with $\Delta\beta$ (4) and applies it to the field via matrix multiplication.

3.4 Panel Descriptions

Panel A: Phase vs. length (Agrawal Eq. 4.1.2)

- **Black line:** Analytic phase $\phi = \Delta\beta \cdot L/2$ computed from $\Delta n = 0.87 \times 10^{-5}$ at $T = 25\,°\mathrm{C}$, $\lambda = 1550\,\mathrm{nm}$.
- **Red dots:** Phase of the simulated Jones matrix element J_{00} measured from `apply_birefringence`.
- **Finding:** Phase increases linearly with length at $35.3\,\mathrm{rad\,m^{-1}}$. Phase error is $0.00e{+}00$ rad (exact match). Beat length $L_B = 178.2\,\mathrm{mm}$.

Panel B: Temperature sensitivity

- **Black line:** Analytic phase $\phi = \Delta\beta(T) \cdot L/2$ at each temperature, where $\Delta\beta(T) = 2\pi\big(\Delta n_{T_0} + c_T(T - 25)\big)/\lambda$.
- **Orange dots:** Phase of simulated J_{00} at each temperature.
- **Finding:** Phase decreases linearly from $-20\,°\mathrm{C}$ to $60\,°\mathrm{C}$. Error is $0.00e{+}00\,\mathrm{rad}$. Temperature coefficient $c_T = -5 \times 10^{-7}\,°\mathrm{C}^{-1}$ confirmed.

Panel C: Bend-induced birefringence (Ulrich Eq. 1)

- **Black line:** Analytic phase $\phi = \Delta\beta(R) \cdot L/2$, including the Ulrich bend term $0.135\,(r_{\text{fiber}}/R)^2$.
- **Green dots:** Phase of simulated J_{00} at each bend radius.
- **Finding:** Phase increases sharply as bend radius decreases. At $R = 2$ mm, $\phi \approx 2.9$ rad; at $R = 20$ mm, $\phi \approx 0.2$ rad. Error is $0.00e{+}00$ rad. Ulrich model confirmed.

Panel D: Δn vs. bend radius (Ulrich Eq. 1)

- **Green dots:** Birefringence Δn extracted from the simulated Jones matrix via the ratio method (two closely spaced fiber lengths, taking the phase difference).
- **Black dashed line:** Ulrich theory: $\Delta n = \Delta n_{T_0} + 0.135\,(r_{\text{fiber}}/R)^2$.
- **Finding:** Extracted Δn falls exactly on the Ulrich theory curve. Maximum extraction error is 0.0000 %.

Panel E: Beat length vs. wavelength

- **Purple line:** Beat length $L_B = \lambda/\Delta n$ (Eq. 6) as a function of wavelength.
- **Gray dashed lines:** Reference markers at 1550 nm and 1310 nm.
- **Finding:** At 1550 nm, $L_B = 178.16$ mm; at 1310 nm, $L_B = 150.57$ mm. Consistent with standard single-mode fiber (long L_B indicates weak birefringence).

Panel F: Validation summary

All five tests (power conservation, phase vs. length, temperature, bend, wavelength) pass with zero error.

4 Chromatic Dispersion

4.1 Source Literature

Agrawal, G. P., *Fiber-Optic Communication Systems*, 5th ed., Wiley, 2021. Section 2.4, Equation (2.4.11), Figure 2.6.

4.2 Theory

A Gaussian pulse with initial width T_0 broadens as it propagates through a dispersive fiber. Per Agrawal [2] Eq. (2.4.11):

$$\sigma(z) = \frac{T_0}{\sqrt{2}} \sqrt{1 + \left(\frac{z}{L_D}\right)^2} \tag{7}$$

where the dispersion length is

$$L_D = \frac{T_0^2}{|\beta_2|} \tag{8}$$

and the group-velocity dispersion (GVD) parameter is

$$\beta_2 = -\frac{D\,\lambda^2}{2\pi\, c} \tag{9}$$

with D the dispersion parameter in $\text{ps}\,\text{nm}^{-1}\,\text{km}^{-1}$.

The CD transfer function in the frequency domain is

$$H(\omega) = \exp\left(-j\,\frac{\beta_2\,\omega^2\,L}{2}\right) \tag{10}$$

4.3 Model

`apply_cd(E, dt, L, wavelength)` applies the transfer function (10) in the frequency domain via FFT.

4.4 Panel Descriptions

Panel A: Gaussian pulse broadening (Agrawal Fig. 2.6)

- **Four curves:** Normalized pulse intensity $|E(t)|^2/\max$ at $z/L_D = 0.0$ (blue), 0.5 (green), 1.0 (orange), and 2.0 (red).
- **Legend:** Analytic RMS width σ for each distance: 21.2 ps, 23.7 ps, 30.0 ps, 47.4 ps.
- **Finding:** The Gaussian pulse broadens symmetrically. At $z/L_D = 0$, $\sigma = 21.2$ ps; at $z/L_D = 2.0$, $\sigma = 47.4$ ps (2.24×). Shape is preserved, consistent with Agrawal Fig. 2.6.

Panel B: RMS width growth ($D = 14.0$ ps/nm/km)

- **Black line:** Analytic curve $\sigma(z)/\sigma_0 = \sqrt{1 + (z/L_D)^2}$.
- **Red dots:** Measured RMS width ratio from `apply_cd`.
- **Red labels:** Relative error at $z/L_D = 0.0$, 0.5, 1.0, 2.0 (all 0.0000 %).
- **Finding:** The measured ratio follows the analytic curve exactly. Growth from 1.0 (at $z = 0$) to ∼2.24 (at $z = 2L_D$). Maximum error $\sim 3.0 \times 10^{-14}$ %.

Panel C: Residual

- **Red line:** Percentage error across the full distance sweep.
- **Black dots:** Test points at $z/L_D = 0.0$, 0.5, 1.0, 2.0.
- **Finding:** Maximum relative error $\sim 3.0 \times 10^{-14}$ % (machine precision). The simulation matches the analytic Gaussian broadening formula exactly.

Panel D: CD transfer function phase (Agrawal Eq. 2.4.11, $z = 2L_D$)

- **Black line:** Unwrapped analytic phase $\phi(f) = -\beta_2(2\pi f)^2 L/2$ at $z = 2L_D$.
- **Red dotted line:** Unwrapped phase of the simulated transfer function $H_{\text{sim}} = \text{FFT}(E_{\text{out}})/\text{FFT}(E_{\text{in}})$.
- **Blue dashed line:** Reference phase at $z = L_D$ (half the test distance).
- **Finding:** Analytic and simulated phases overlap perfectly. The phase is quadratic in frequency (parabolic), the signature of chromatic dispersion. At the Nyquist frequency (0.5 THz), the accumulated phase is ∼1200 rad.

Panel E: Material and waveguide dispersion (Hui [2], Keck [3])

- **Blue line:** Material dispersion $D_{\mathrm{mat}}(\lambda)$ from the approximate Sellmeier model.
- **Orange line:** Waveguide dispersion $D_{\mathrm{wg}} = -3.0\,\mathrm{ps/nm/km}$ (constant).
- **Black dashed line:** Total dispersion $D_{\mathrm{total}} = D_{\mathrm{mat}} + D_{\mathrm{wg}}$.
- **Gray dashed lines:** Reference markers at 1550 nm and 1310 nm.
- **Finding:** At 1550 nm, $D_{\mathrm{total}} = 17.0 + (-3.0) = 14.0\,\mathrm{ps/nm/km}$ (standard SMF-28). Total dispersion crosses zero near 1310 nm.

Panel F: Validation summary

$D_{\mathrm{total}} = 14.0\,\mathrm{ps/nm/km}$, $\beta_2 = -1.786\times10^{-26}\,\mathrm{s^2\,m^{-1}}$, $L_D = 50.40\,\mathrm{km}$, max error $= 3.0\times10^{-14}\,\%$. Gaussian broadening matches analytic to $< 0.001\,\%$.

5 Polarization Mode Dispersion

5.1 Source Literature

Razavi, B., *Design of Integrated Circuits for Optical Communications*, 2nd ed., Wiley, 2012. Section 2.5, Figure 2.11.

5.2 Theory

PMD causes the differential group delay (DGD) between two principal states of polarization to follow a Maxwellian distribution:

$$p(\tau) = \frac{2}{\sqrt{\pi}} \frac{\tau^2}{\sigma_m^3} \exp\left(-\frac{\tau^2}{\sigma_m^2}\right) \tag{11}$$

where $\tau = \mathrm{DGD}$ and the Maxwellian scale parameter is

$$\sigma_m = \frac{D_{\mathrm{PMD}}\sqrt{L}}{\sqrt{3}} \tag{12}$$

with D_{PMD} the PMD coefficient. The RMS DGD is

$$\mathrm{DGD}_{\mathrm{rms}} = D_{\mathrm{PMD}}\sqrt{L} \tag{13}$$

5.3 Model

`apply_pmD(E, dt, L, pm_dispersion)` samples a Maxwellian DGD and applies a frequency-domain phase shift between the two polarization components.

5.4 Panel Descriptions

Panel A: DGD distribution (Razavi Fig. 2.11)

- **Blue histogram:** Distribution of DGD values from 5000 realizations of `apply_pmD` at $L = 100\,\mathrm{km}$.
- **Black line:** Analytic Maxwellian PDF (Eq. 11) with RMS DGD $= 31.6\,\mathrm{ps}$.
- **Gray dashed line:** Expected RMS DGD $= 31.6\,\mathrm{ps}$.

- **Blue dotted line:** Measured mean DGD = 29.20 ps.
- **Finding:** The simulated DGD histogram matches the analytic Maxwellian distribution. KS test $p = 0.45$ (null hypothesis not rejected). Measured mean (29.20 ps) and RMS (31.74 ps) match expected values (29.14 ps, 31.62 ps) within statistical fluctuation.

Panel B: QQ-plot (Maxwellian fit)

- **Blue dots:** Sample quantiles of the simulated DGD vs. theoretical Maxwellian quantiles.
- **Black dashed line:** Ideal $y = x$ line (perfect agreement).
- **Finding:** Correlation $r = 0.9998$. The simulated DGD distribution is Maxwellian across the full range (0 to 80 ps).

Panel C: Histogram residual

- **Red bars:** Difference between measured histogram density and analytic Maxwellian PDF at each bin.
- **Finding:** Residuals randomly distributed around zero with no systematic pattern. Maximum residual $\sim$0.003, consistent with statistical noise from 5000 realizations.

Panel D: PMD scaling DGD $\sim \sqrt{L}$ (Razavi Sec. 2.5)

- **Green dots:** RMS DGD measured from `apply_pmD` at $L = 10, 25, 50, 75, 100, 150, 200$ km.
- **Black dashed line:** Linear fit: $\mathrm{DGD}_{\mathrm{rms}} = 3.1662\,\sqrt{L}$ ps.
- **Gray dotted line:** Nominal prediction: $\mathrm{DGD}_{\mathrm{rms}} = 3.1623\,\sqrt{L}$ ps.
- **Finding:** RMS DGD scales linearly with $\sqrt{L}$, confirming the fundamental PMD scaling law (Eq. 13). Fitted PMD coefficient = 3.1662 ps/$\sqrt{\mathrm{km}}$ vs. nominal 3.1623 ps/$\sqrt{\mathrm{km}}$. $R^2 = 0.99993$.

Panel E: PMD coefficient consistency

- **Purple dots:** Extracted PMD coefficient at each length: $D_{\mathrm{PMD}} = \mathrm{DGD}_{\mathrm{rms}}/\sqrt{L}$.
- **Gray dashed line:** Nominal PMD coefficient = 0.100 ps/$\sqrt{\mathrm{km}}$.
- **Finding:** The extracted PMD coefficient is approximately constant across all fiber lengths (10 to 200 km), confirming consistent PMD application regardless of distance.

Panel F: Validation summary

Mean DGD = 29.205 ps (expected 29.135 ps), RMS DGD = 31.736 ps (expected 31.623 ps), KS $p = 0.45$ (not rejected), PMD coefficient = 3.1662 ps/$\sqrt{\mathrm{km}}$ (expected 3.1623), $R^2 = 0.99993$.

References

[1] G. Keiser, *Optical Fiber Communications*, 5th ed., McGraw-Hill, New York, 2015.

[2] G. P. Agrawal, *Fiber-Optic Communication Systems*, 5th ed., Wiley, Hoboken, NJ, 2021.

[3] R. Ulrich, S. C. Rashleigh, and W. Eickhoff, "Bending-induced birefringence in single-mode fibers," *Opt. Lett.*, vol. 5, no. 6, pp. 273–275, 1980.

[4] R. S. Hui and M. O'Sullivan, *Fiber Optic Measurement Techniques*, Academic Press, 2009.

[5] B. Razavi, *Design of Integrated Circuits for Optical Communications*, 2nd ed., Wiley, Hoboken, NJ, 2012.

[6] D. B. Keck, "Fundamentals of integrated fiber optics," in *Integrated Optics*, T. Tamir, Ed., Springer, 1981, pp. 173–230.